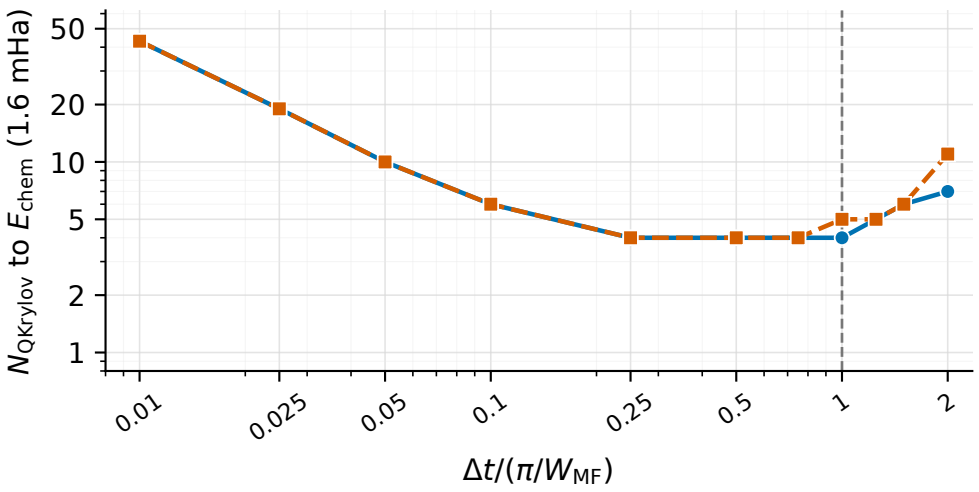
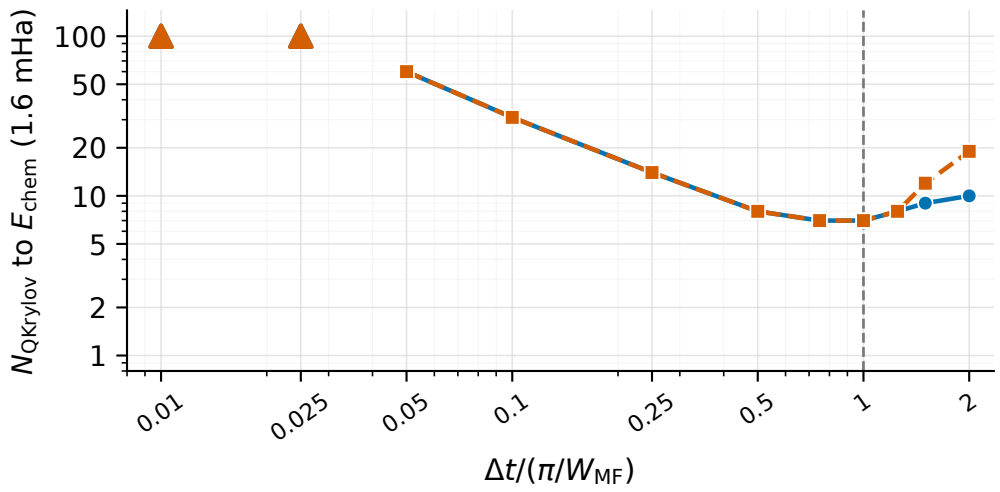
QRTE timestep scan: exact vs second-order Trotter

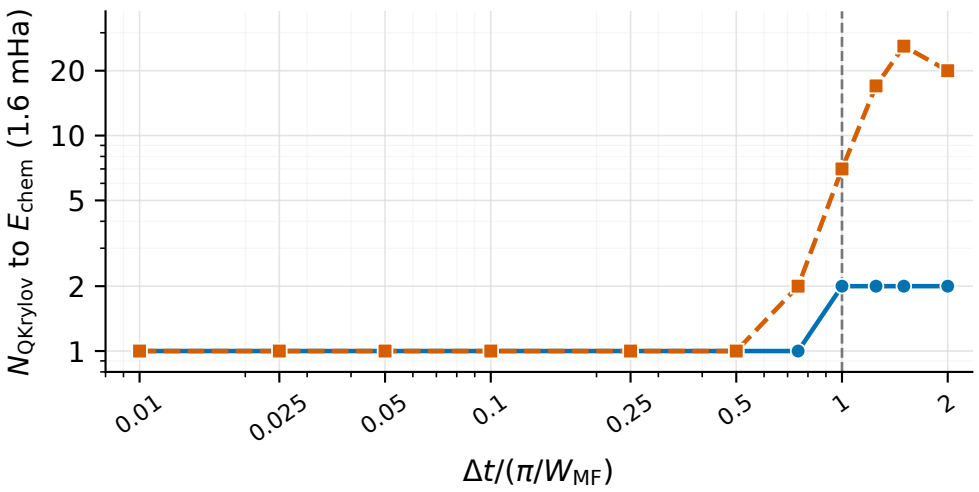
N₂/cc-pVTZ, spin 0
$\pi/W_{\mathrm{MF}} = 0.653130$ a.u.



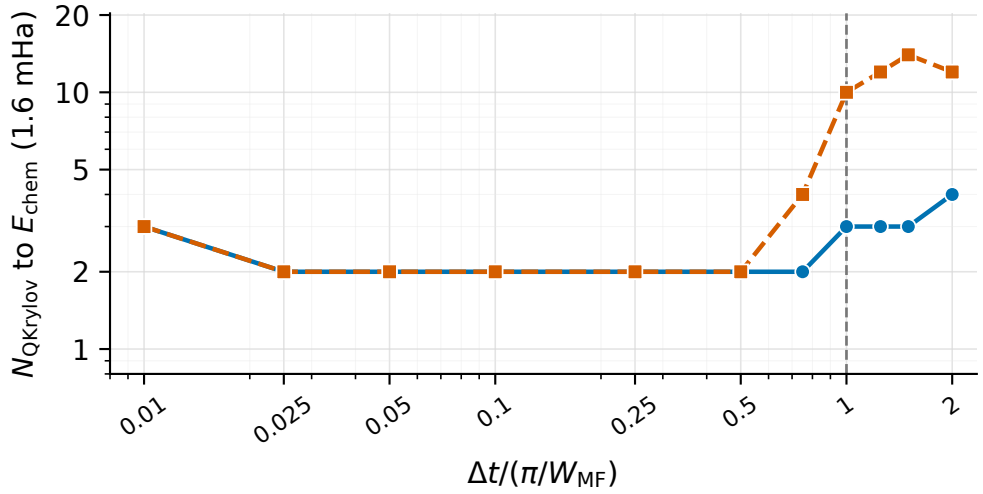
N₂/cc-pVTZ, spin 2
$\pi/W_{\mathrm{MF}} = 0.932281$ a.u.



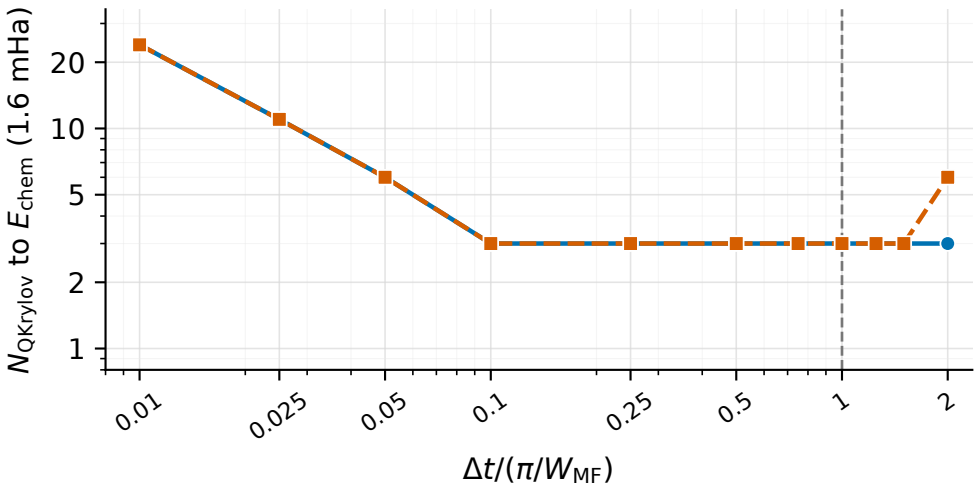
BeH₂/cc-pVTZ, spin 0
$\pi/W_{\mathrm{MF}} = 1.327096$ a.u.



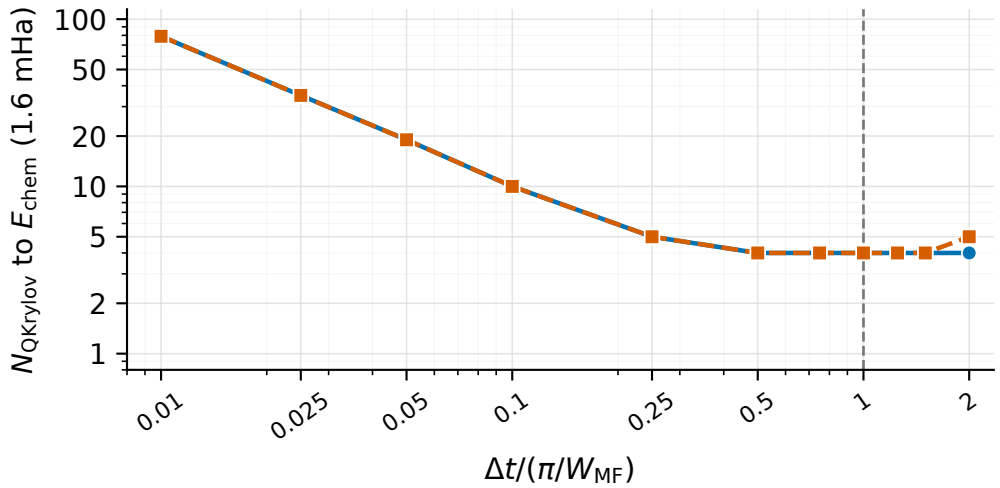
BeH₂/cc-pVTZ, spin 2
$\pi/W_{\mathrm{MF}} = 1.717979$ a.u.



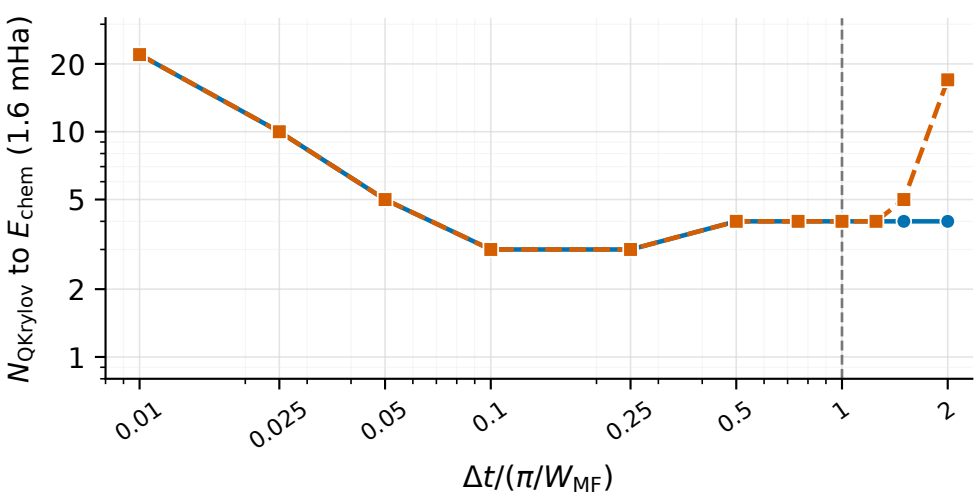
H4/STO-3G, spin 0
$\pi/W_{\mathrm{MF}} = 0.724457$ a.u.



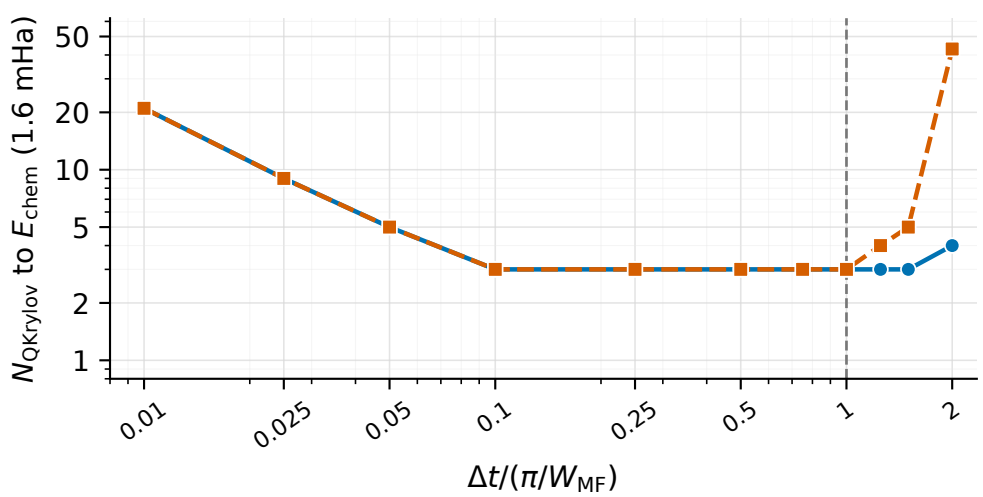
H6/STO-3G, spin 0
$\pi/W_{\mathrm{MF}} = 0.461237$ a.u.



H6/cc-pVTZ, spin 0
$\pi/W_{\mathrm{MF}} = 0.699023$ a.u.



H6/cc-pVTZ, spin 2
$\pi/W_{\mathrm{MF}} = 0.887815$ a.u.



Exact exponential QRTE
QRTE: second-order Trotter, one slice
$\Delta t = \pi/W_{\mathrm{MF}}$
△ No hit within 100 iterations (series color)